THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

MECON 6015 Economics of AI and Innovation

GENERAL INFORMATION

Instructor: Prof. Yanhui Wu

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Consultation times: Tuesday, 4-5 pm

TAs: Yele Ma, Yuhao Zhang

Pre-requisites: Intermediate microeconomics and macroeconomics; Elementary (college level) calculus and econometrics

COURSE DESCRIPTION

This course provides rigorous analysis of how the advances in artificial intelligence and digital technology transform the economy and society. It will not only cover conventional topics on the impact of AI on innovation, business strategy, firm dynamics, employment, and economic growth, but also guide students to explore frontier topics related to social media, digital entertainment, gig economy, and algorithmic regulation. The course will be grounded on economic analysis and empirical methods with an emphasis on applications instead of methodological sophistication. Students are expected to master basic analytic and empirical tools to study knowledge, innovation, and AI in economics and management. Students will be assessed based on problem-solving assignments, a business analytical project, and a final exam.

COURSE OBJECTIVES

1. Provide students with basic analytic tools for the study of ideas, knowledge, innovation, and AI from the economics perspective.
2. Increase students' awareness of the economic and social impacts of recent advances of AI and digital technology
3. Develop students' ability to apply economics to real-world phenomena, business strategy, and policy analysis both theoretically and empirically
4. Enhance students' ability of critical thinking

Programme Learning Outcomes

PLO1: Understanding of fundamental theories and new development in economics

PLO2: Mastering of skills in analyzing economic data

PLO3: Demonstration of ability to apply economic knowledge and analytical skills to address policy and business problems with a global outlook

PLO4: Awareness of ethical concerns in economic issues

PLO5: Mastering of communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	Aligned Programme Learning Outcomes
CLO1 Gain a solid understanding of the fundamental economic problems associated with knowledge and artificial intelligence.	PLO 1, PLO 4, PLO5

CLO2 Understand the business opportunities and challenges raised by advances in AI and digital technology.	PLO 1, PLO 2, PLO 4, PLO 5
CLO3 Be familiar with economic analysis of information, innovation, and technological progress.	PLO 1, PLO 2, PLO 3, PLO 5
CLO4 Learn how to apply theoretical models and empirical methods to solve real economic and business problems.	PLO 1, PLO 2, PLO 3, PLO 5

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	Expected contact hour	Study Load (% of study)
T&L1. **Lectures**	30 hours	25%
T&L2. **Reading course material including research papers**	42 hours	35%
T&L3. **Preparing for homework assignments, projects, and exams.**	36 hours	30%
T&L4. **Group discussion**	12 hours	10%
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Total	120 hours	100%

Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. Class engagement	Attendance and class participation	10 %	CL01 CL02 CL03 CL04
A2. Homework assignment	Two problem sets	40 %	CL01 CL02 CL03 CL04
A3. Final exam	Closed-book written exam	25 %	CL01 CL02 CL03 CL04
A4. Final project report	Group project (research proposal or business analytics report)	25 %	CL01 CL02 CL03 CL04
	Total	100%	

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	Strong evidence of superb ability to fulfill the intended learning outcomes of the course at all levels of learning: describe, apply, implement, evaluate and synthesis.
B+, B, B-	Strong evidence of ability to fulfill the intended learning outcomes of the course at all levels of learning: describe, apply, implement, evaluate and synthesis.
C+, C, C-	Evidence of adequate ability to fulfill the intended learning outcomes of the course at low levels of learning; such as describe and apply, but not at high levels of learning such as evaluate and synthesis.
D+, D	Evidence of basic familiarity with the subject.
F	Little evidence of basic familiarity with the subject.

Assessment Rubrics for Each Assessment (Please provide us the details in a separate file if the space here is not enough)

Please see the separate document for details.

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

Part I. Fundamentals (classes 1-6)

1.1 Digital transformation and the AI awakening
1.2 Economics of ideas and knowledge
1.3 Innovation and creative destruction
1.4 AI, management, and business strategy
1.5 AI, employment, and income distribution

Part II. Advanced topics (classes 7-10)

2.1 Social media
2.2 Digital entertainment
2.3 The gig economy
2.4 Algorithmic regulation

Part III. Review and Discussions (classes 11-12)

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

There is no required textbook for this course. A set of slides and reading material (including academic papers and business articles) will be distributed to students. A list of business bestsellers related to the topics will be recommended to students from time to time.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

○ conducting mid-term survey in additional to SFTL around the end of the semester

○ Online response via Moodle site

✓ Others: ___SFTL_____________ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

1. This is an active learning course, and attendance and participation are extremely important. Please observe appropriate classroom etiquette and be considerate to others. In particular, laptop use should be limited to course-related activities, and cell phones are not allowed in class.

2. Students are encouraged to work together in groups to solve the problem sets. However, each student must turn in his or her own homework. Copying another student's answers is not permitted even with consent. All assignments including the term project report must be typewritten.

3. Plagiarism and cheating in exams are serious academic offenses.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)